

MATHEUS LAJARIN ZEM

Mid-level Full Stack Developer · Java · Spring · Angular

Piracicaba/SP · +55 (19) 99985-1929 (WhatsApp) · matheuslajazem@gmail.com
linkedin.com/in/matheus-lajarin-zem-17b39b289 · github.com/MatheusZem12

PROFESSIONAL SUMMARY

Full Stack Developer working end to end: data modeling, APIs, microservices, frontend, CI/CD and production operations. Corporate experience in the agribusiness ecosystem (TOTVS PIMS ERP), across two microservice product lines (Spring Boot 3, Spring Cloud Gateway, Eureka, OpenFeign) with mission-critical integrations — SAP, TOTVS Protheus, B2B platforms and industrial hardware — on Oracle, SQL Server and PostgreSQL databases.

In parallel, I design, build and **operate my own products in production**: a Linux VPS hosts three Spring Boot backends in Docker, with a GitHub Actions → GHCR → Watchtower pipeline (pull-based deployment), Cloudflare Tunnel at the edge, and real DEV/STAGING/PROD environments. Focus on verifiable engineering: versioned Flyway migrations, testing with JUnit, Testcontainers, Playwright and Vitest, and security with JWT, OAuth2, 2FA, AES-GCM encryption and rate limiting.

Knowledge of messaging (RabbitMQ) and hands-on use of asynchronous patterns: domain events, polling-based task queues, scheduled jobs and virtual threads. I use AI assistants (Claude and KIMI) as pair programmers — coding, review, debugging and architecture — always validating the results.

PROFESSIONAL EXPERIENCE

TOTVS (agribusiness sector) — Mid-level Full Stack Developer (Contractor) *Nov 2024 – present*

Development and maintenance of agribusiness management platforms — costs, cash flow, materials, suppliers, weighing/scales, industrial production and harvest optimization — in a microservices architecture integrated with the PIMS ERP.

- **Microservices:** work on two product lines totaling ~30 Spring Boot 3 services (Java 17/21), with two gateway patterns — Spring Cloud Gateway with static routes + OpenFeign in one line, and Gateway + Netflix Eureka (service discovery, 13 routes) in the other; observability with Actuator and Prometheus.
- **Mission-critical integrations:** SAP connector (contracts, invoices, weighings, provisions and costs), PIMS ↔ TOTVS Protheus interfaces, CIB2B procurement platform, Milvus

helpdesk (REST API + RPA with Selenium) and Toledo industrial scale reading over raw TCP socket.

- **Multi-database:** PostgreSQL in the products, Oracle in the PIMS ERP and SQL Server in PIMS Multicultivo — data modeling, query optimization and scheduled synchronization jobs between ERP and products.
- **Legacy modernization:** rewriting Java 8 / Spring Boot 2 integrations (WAR on Tomcat) to Java 17/21 / Spring Boot 3, and contributing to the evolution of a legacy product (Angular 8 → Angular 19).
- **Frontend:** Angular 17–20 portals (PrimeNG, TailwindCSS, ECharts/Chart.js, i18n), including Angular 20 with SSR; maintenance of a livestock management product in Python/Django REST + Vue 3 supporting Oracle, SQL Server and PostgreSQL simultaneously.
- **CI/CD:** GitHub Actions pipelines deploying the portal to Azure Static Web Apps across 3 environments; DEV/STAGING/PROD Git flow.

CNH Industrial — Project Management Intern *Jun 2022 – Jun 2024*

Parts management in projects, from registration to supplier relations; automation of data control and analysis with macros and pivot tables.

Elring Klinger — Kaizen Intern *Mar 2022 – May 2022*

Follow-up of Kaizen requests and automation of report generation for several departments.

PERSONAL PROJECTS

Complete products that I conceived, developed and operate alone in production — backend, frontend, database, CI/CD and infrastructure — using AI (Claude and KIMI) as a pair programmer.

Monevas — personal finance and investment analysis *hmg.monevas.com*

- **Backend:** Java 21 + Spring Boot as a modular Maven monolith with 8 modules and ~120 Flyway migrations (Hibernate in validate mode); JWT + e-mail 2FA, AES-GCM encryption of sensitive data at rest, custom rate limiting with trusted proxies and anti-slowloris protections.
- **Market data ingestion:** Python service (FastAPI) collecting from B3, CVM and Yahoo Finance — quotes, dividends, financial statements and REITs — into a database with 8.9 million candles; global throttling and retry with backoff against rate limits.

- **Analytics:** native desktop application (PySide6 + pyqtgraph) with a 4-layer architecture enforced by tests; 5 valuation methods, screener and percentile-based scoring; authentication delegated to the backend.
- **Frontend:** Angular 21 + PrimeNG, i18n (pt/en/es), Vitest tests as a build gate, deployed on Vercel.
- **Infra/CI:** GitHub Actions → GHCR → Watchtower on the VPS (pull-based deployment, GitHub never reaches the server); Cloudflare Tunnel at the edge; a real staging environment before promoting to production.
- **Challenges solved:** JVM OOMKill (exit 137) on a 4 GB VPS during price synchronization, fixed with per-container memory limits and MaxRAMPercentage tuning; yfinance dependency pinned after columns silently became NULL; diagnosis of a silent Watchtower restart loop (Docker API incompatibility) that was blocking deployments.

Lingua — language learning with spaced repetition

- Backend in Java 21 + Spring Boot + PostgreSQL; desktop (Electron/Wayland) and Android (Capacitor) clients; speech synthesis (edge-tts) and speech recognition (Whisper) running on CPU inside the container.
- **Spaced-repetition engine ported from JavaScript to Java** with differential tests (200 trajectories compared bit by bit against the original engine); parameters chosen through synthetic-student simulation with ablation — 7-day retention of 81% vs 7% in the worst configuration.
- Multi-provider AI integration on the client (Anthropic, OpenAI, Gemini) with structured output; lyrics and metadata lookup (LRCLIB, Deezer) with candidates fetched in parallel using virtual threads.
- **Challenges solved:** a silent uid permission failure on a Docker volume turned into a build error through smoke tests baked into the image; synchronization of thousands of audio files across devices via catalog, manifest and reconciliation.

Monitorame — logistics: fleets, shipments, sensors and routes

- **Modular Spring Boot 3.5 monolith with 15 Maven modules** and an acyclic dependency graph; inter-module communication through Spring domain events; soft delete across all resources.
- Routing with OSRM (distance matrix + routes), deterministic haversine fallback and stop optimization with nearest-neighbor + 2-opt; Angular 22 + Leaflet frontend.

- **Quality:** CI with 3 jobs — backend with Testcontainers (real PostgreSQL), frontend, and Playwright e2e; 3-stage compose with chained healthchecks and Nginx proxying /api (eliminating CORS and API URLs in the bundle).

Activity Manager — time tracking with AI classification (Linux)

- Electron integrated with the Hyprland/Wayland compositor (global shortcuts and window/idle collection via the event socket); Spring Boot + PostgreSQL backend.
- **4-stage classification** (idle → user rules → cache → AI queue): the server enqueues tasks and devices poll outbound (works behind NAT), running the model locally (Ollama/Claude); categories guaranteed by an enum in the structured-output schema; the server never sees window titles — privacy by architecture.

Educa — multi-role academic portal

Spring Boot 3.5 + PostgreSQL + Flyway; Angular 21 + PrimeNG + Tailwind; dashboards (Chart.js) for students, teachers and institutions across 23 screens; JWT; 3 environments.

Other projects

- **mail-reader** — desktop e-mail client (Electron) with Gmail API and Microsoft Graph via OAuth; tokens encrypted in the OS keychain.
- **Farol** — self-hosted family location sharing: Node/Express + Socket.IO + PostgreSQL/PostGIS (geofencing with ST_DWithin), Leaflet PWA, OwnTracks integration.
- **git-manager** — desktop Git client (Electron + simple-git) with commit graph and visual staging.
- **terminal-animado** — published VS Code extension; animations optimized through real CPU measurements (62% → ~1% of a core by moving everything to GPU-composited transform/opacity).
- **Health observatory** — public data pipeline over DATASUS (SIM/SINASC/SIH/CNES): Python + DuckDB/Parquet, decoding of the proprietary .DBC format and epidemiological SQL analyses.

TECHNICAL SKILLS

- **Languages:** Java (advanced) · TypeScript/JavaScript · Python · SQL · Go · Dart
- **Backend:** Spring Boot 2/3 · Spring Cloud (Gateway, Eureka, OpenFeign) · Spring Security (JWT, OAuth2) · REST APIs · JPA/Hibernate · FastAPI · Django REST · Node/Express · Socket.IO/WebSocket

- **Messaging & async:** RabbitMQ · Spring domain events · polling-based task queues · @Scheduled/cron · virtual threads (Java 21)
- **Frontend:** Angular 17–22 (PrimeNG, TailwindCSS, RxJS, SSR) · Vue 3 · Electron · Flutter · Capacitor
- **Databases:** PostgreSQL (+ PostGIS) · Oracle · SQL Server · data modeling and query optimization · Flyway · DuckDB/Parquet
- **DevOps & Infra:** Docker/Compose · GitHub Actions · GHCR · Watchtower · Cloudflare Tunnel · Nginx · Linux/VPS · Vercel · Azure Static Web Apps · Prometheus/Actuator
- **Testing:** JUnit · Testcontainers · Playwright (e2e) · Vitest · differential and load testing
- **Applied AI:** LLM integration (Anthropic, OpenAI, Gemini) with structured output · RPA (Selenium) · AI as a pair programmer (Claude, KIMI)

EDUCATION

Computer Engineering — Escola de Engenharia de Piracicaba (EEP), Brazil · expected graduation: Dec 2026 (evening program).

LANGUAGES

Portuguese — native. **English** — advanced (Red Balloon, Cultura Inglesa and Wise Up courses; hands-on experience traveling in the US).